Ethical Analysis HealthSys

By: Mert Affan, Katrine Risto, Hao Tang

Issue: **Unclear intended use and lack of transparency**
If the intended use as well as user experience is not clear, unsuitable users might have a bad experience with the app. For instance, if the instructions for intended use do not clearly state that the app is for educational purposes only and that all information provided is merely a recommendation, the absence of user health data may result in advice that is less tailored and potentially less effective for the user. Also, leaving out warnings and precautions can also lead to a bad user experience and have a negative effect on the users physical and mental health.

Solution
Form a clear instruction of intended use as well as a description of what should be expected by the app. The intended use should include the limitations of the app and of wanted results by the user, setting reasonable expectations for the user. As a complementary instruction, the user should be given warnings of possible effects when the app is used incorrectly or with any underlying health issue. It should be clear if the app is not suitable for users with underlying health issues or with other diagnoses. It should also state that if there are some unclearities whether the app is suitable for the user or not, the user should be advised by a doctor first.

Issue: Data Security and Risk of Breach
Health-related data is highly sensitive, and unauthorized access to or disclosure of such data could lead to discrimination and emotional distress for users.

Solution:
Direct storage and transmission of users' real data should be avoided. Data encryption and anonymization methods should be implemented to ensure compliance with data protection regulations, and data access should be limited to authorized personnel.

Issue: AI Accuracy
Since the questions are generated by and a LLM they have a risk of being incorrect, outdated or misleading. Users might therefore take wrong advice or learn incorrect behaviour. Health is a living subject and some information that was correct last year might be incorrect this year.

Solution:
All of the questions should be fact checked continuously and they should also be restricted to follow the guidelines of “Livsmedelsverket”.

Issue: Manipulation
We want the user to want to come back to the app everyday. To achieve this we will implement streaks, quizzes and rewards in the app. However, streaks induce stress and could potentially give the user bad conscience. Rewards and streaks also have the possibility to generate addictive behaviour.

Solution:
The streaks should be easily recovered or have different freezing options, taking “Duoling” as an example. Positive feedback is also good to have. For example, “welcome back! Do you want to start where we left off?”. The user should also be able to adjust their frequency of use, with a usage alert. For example the user could be reminded every 5 minutes to take a break.

Issue: Bias & Inclusivity
LLMs rely on large amounts of internet-based data, which can introduce bias. For instance, much of the available information is Western centered, meaning the generated questions may unintentionally exclude or misrepresent people from other cultural or dietary traditions. Furthermore, our app may not be suitable for certain groups, such as very elderly users or those with visual impairments, which exacerbates inequalities in the digital health field.

Solution:
Since food is important for some traditions the questions should not favour any specific area. They should be personalized to the user. We should create a diverse question dataset, and generate questions based on the user’s habits. If he or she only eats asian food, the questions should also be based on asian foods. Furthermore, LLM should be prohibited from generating stigmatizing and judgmental language.
When designing an app, the interface should be simple rather than overly complex, using clear and easy-to-understand non-technical language, and may include integrated voice prompts. Conduct usability testing for different user groups to ensure that all users have equal access to healthy learning opportunities.